

Question Bank- Part 2

MCQs with Answers

1. **Cryptography is the science of:**
 - a) Building secure networks
 - b) Secret or hidden writing
 - c) Data transmission
 - d) Computer hardware security**Answer: b) Secret or hidden writing**
2. **Which of the following is a type of substitution cipher?**
 - a) AES
 - b) Caesar Cipher
 - c) Diffie-Hellman
 - d) RSA**Answer: b) Caesar Cipher**
3. **In symmetric encryption, the key for encryption and decryption are:**
 - a) Different
 - b) Public
 - c) The same
 - d) Based on hash values**Answer: c) The same**
4. **Asymmetric encryption uses how many keys?**
 - a) One
 - b) Two
 - c) Three
 - d) None**Answer: b) Two**
5. **Which algorithm is considered the first key agreement algorithm, enabling secure key exchange without prior shared secrets?**
 - a) RSA
 - b) AES
 - c) Diffie-Hellman
 - d) DES**Answer: c) Diffie-Hellman**
6. **A hash function is best described as:**
 - a) A two-way encryption algorithm
 - b) A function that maps data of any size to a fixed size
 - c) A method for symmetric key exchange
 - d) A type of firewall**Answer: b) A function that maps data of any size to a fixed size**

Question Bank- Part 2

7. Which hash function produces a 256-bit fixed-length output?

- a) MD5
- b) SHA-1
- c) SHA-256
- d) HMAC

Answer: c) SHA-256

8. HMAC (Hash-based Message Authentication Code) is used to verify:

- a) Only data integrity
- b) Only data authenticity
- c) Both data integrity and authenticity
- d) Encryption strength

Answer: c) Both data integrity and authenticity

9. The main goal of a Cryptanalysis attack is to:

- a) Recover the plaintext of a single message
- b) Recover the key in use
- c) Disrupt network service
- d) Alter message content

Answer: b) Recover the key in use

10. A firewall's default-deny approach, where packets are dropped unless specifically allowed, is known as a:

- a) Blacklist
- b) Whitelist
- c) Grey list
- d) Stateful list

Answer: b) Whitelist

11. Which generation of firewalls introduced stateful inspection?

- a) 1st Generation
- b) 2nd Generation
- c) 3rd Generation
- d) Next-Generation

Answer: b) 2nd Generation

12. A firewall that acts as an intermediary, terminating client connections and initiating new ones to the destination, is a:

- a) Packet-filtering Firewall
- b) Stateful Inspection Firewall
- c) Proxy Firewall
- d) WAF

Answer: c) Proxy Firewall

Question Bank- Part 2

13. A Web Application Firewall (WAF) primarily operates at which OSI layer?

- a) Layer 3 (Network)
- b) Layer 4 (Transport)
- c) Layer 7 (Application)
- d) Layer 2 (Data Link)

Answer: c) Layer 7 (Application)

14. A UTM (Unified Threat Management) appliance typically integrates all EXCEPT:

- a) Firewall
- b) Intrusion Prevention System (IPS)
- c) Antivirus
- d) Physical door locks

Answer: d) Physical door locks

15. A network segment that acts as a buffer zone between an internal network and the untrusted internet is called a:

- a) VPN
- b) DMZ
- c) LAN
- d) WAN

Answer: b) DMZ

16. Which VPN type is used to connect the networks of different office locations for the same company?

- a) Remote Access VPN
- b) Intranet-based Site-to-Site VPN
- c) Extranet-based Site-to-Site VPN
- d) Peer-to-Peer VPN

Answer: b) Intranet-based Site-to-Site VPN

17. The protocol that provides security for Internet Protocol communication by verifying sessions and encrypting data packets is:

- a) PPTP
- b) L2TP
- c) IPSec
- d) SSL

Answer: c) IPSec

18. WAP (Wireless Application Protocol) was designed primarily for:

- a) High-bandwidth desktop computers
- b) Mobile devices with limitations
- c) Fiber-optic networks
- d) Satellite communication

Answer: b) Mobile devices with limitations

Question Bank- Part 2

19. The security layer in the WAP protocol stack, based on TLS, is called:

- a) WSP
- b) WTP
- c) WTLS
- d) WDP

Answer: c) WTLS

20. PGP (Pretty Good Privacy) provides services for:

- a) Only email confidentiality
- b) Email confidentiality and authentication
- c) Web browsing security
- d) Firewall configuration

Answer: b) Email confidentiality and authentication

21. The default non-encrypted port for the SMTP protocol is:

- a) Port 25
- b) Port 110
- c) Port 143
- d) Port 443

Answer: a) Port 25

22. Which protocol allows email clients to keep messages on the server and access them from multiple devices?

- a) POP3
- b) IMAP
- c) SMTP
- d) FTP

Answer: b) IMAP

23. The strength of a symmetric encryption algorithm is primarily determined by:

- a) The complexity of the algorithm's name
- b) The size (length) of the key
- c) The speed of encryption
- d) The type of data being encrypted

Answer: b) The size (length) of the key

24. The Vigenère cipher is an example of a:

- a) Stream cipher
- b) Block cipher
- c) Polyalphabetic substitution cipher
- d) Transposition cipher

Answer: c) Polyalphabetic substitution cipher

Question Bank- Part 2

25. Which symmetric encryption algorithm uses a 56-bit key and operates on 64-bit blocks?

- a) AES
- b) RSA
- c) DES
- d) IDEA

Answer: c) DES

26. In public-key cryptography, a message encrypted with the recipient's public key can only be decrypted with the recipient's:

- a) Public Key
- b) Private Key
- c) Session Key
- d) Hash Key

Answer: b) Private Key

27. A major disadvantage of symmetric-key cryptography is:

- a) It is too slow for modern networks
- b) The key must be exchanged securely
- c) It does not provide confidentiality
- d) It cannot be used for authentication

Answer: b) The key must be exchanged securely

28. Cipher Block Chaining (CBC) mode works by:

- a) Encrypting each block independently
- b) XORing the plaintext block with the previous ciphertext block before encryption
- c) Using a stream of key bits
- d) Compressing data before encryption

Answer: b) XORing the plaintext block with the previous ciphertext block before encryption

29. A type of cryptanalytic attack where the attacker has some known plaintext-ciphertext pairs is called:

- a) Ciphertext-Only Analysis (COA)
- b) Known-Plaintext Analysis (KPA)
- c) Chosen-Plaintext Analysis (CPA)
- d) Man-In-The-Middle (MITM)

Answer: b) Known-Plaintext Analysis (KPA)

30. Ingress filtering in a firewall inspects:

- a) Outgoing traffic from the internal network
- b) Incoming traffic from the external network
- c) Only email traffic
- d) Traffic within the DMZ

Answer: b) Incoming traffic from the external network

Question Bank- Part 2

31. A firewall action where a packet is not allowed through and no error message is sent is called:

- a) Accepted
- b) Rejected
- c) Denied
- d) Filtered

Answer: c) Denied

32. A Next-Generation Firewall (NGFW) includes capabilities beyond stateful inspection, such as:

- a) Application awareness and control
- b) Only IP address filtering
- c) Basic packet header inspection
- d) Physical security

Answer: a) Application awareness and control

33. The primary purpose of a VPN is to:

- a) Increase internet speed
- b) Create a private network over a public infrastructure
- c) Block malicious websites
- d) Encrypt local hard drives

Answer: b) Create a private network over a public infrastructure

34. Which VPN protocol is an open-source protocol commonly used for creating secure point-to-point connections?

- a) PPTP
- b) L2TP
- c) IPSec
- d) OpenVPN

Answer: d) OpenVPN

35. In the WAP architecture, the component that typically translates between WTLS and TLS is the:

- a) WAP Device
- b) Web Server
- c) WAP Gateway
- d) Bearer Network

Answer: c) WAP Gateway

36. The protocol in WAP responsible for managing transactions of requests and responses is:

- a) Wireless Session Protocol (WSP)
- b) Wireless Transaction Protocol (WTP)
- c) Wireless Datagram Protocol (WDP)
- d) Wireless Transport Layer Security (WTLS)

Answer: b) Wireless Transaction Protocol (WTP)

Question Bank- Part 2

37. PGP uses which of the following for email compatibility, expanding the message by 33%?

- a) ZIP compression
- b) Radix-64 conversion
- c) Base32 encoding
- d) Binary-to-text encoding

Answer: b) Radix-64 conversion

38. The secure port for POP3 is:

- a) Port 995
- b) Port 993
- c) Port 465
- d) Port 587

Answer: a) Port 995

39. Which characteristic is an advantage of a Stateful Firewall over a Stateless one?

- a) It is faster and uses less memory.
- b) It can track the state of active connections and context.
- c) It is simpler to configure.
- d) It inspects packets in complete isolation.

Answer: b) It can track the state of active connections and context.

40. The "public" key in asymmetric cryptography is used to:

- a) Decrypt messages
- b) Create digital signatures
- c) Encrypt messages
- d) Both b and c

Answer: d) Both b and c (Create digital signatures and Encrypt messages)

Question Bank- Part 2

Incomplete Quotations.

1. Cryptography is the science of _____ writing. **secret or hidden**
2. The two main components of cryptography are encryption and **authentication & integrity**
3. In symmetric encryption, the key for encryption and decryption are the **same**.
4. Asymmetric encryption uses a pair of keys: a public key and a **private** key.
5. The Caesar Cipher is vulnerable to brute-force attacks because its key space is only **25** possible keys.
6. A _____ cipher uses multiple substitution alphabets (e.g., Vigenère cipher). **polyalphabetic**
7. The strength of a symmetric algorithm is determined by the **size** of the key.
8. DES uses a **56-bit** -bit symmetric key.
9. AES is an example of a **block** cipher, encrypting data in blocks.
10. Cipher Block Chaining (CBC) mode XORs plaintext with the previous _____ before **ciphertext** encryption.
11. Diffie-Hellman is the first _____ algorithm, allowing secure key exchange. **key agreement**
12. RSA is a widely used **public** -key algorithm.
13. A hash function maps data of any size to data of a **fixed** size.
14. The fixed-length output of a hash function is often called a hash value or digital **fingerprint**
15. HMAC provides both data integrity and _____ using a shared secret key. **authenticity**
16. Cryptanalysis is the science of **breaking** encryption.
17. A _____ attack intercepts communication between two parties. **man in the middle**
18. A firewall inspecting incoming traffic from the internet is performing **ingress** filtering.
19. The firewall policy approach that blocks all traffic except what is explicitly allowed is called a **whitelist**
20. A _____ firewall maintains a state table to track active connections. **Stateful Inspection**
21. A _____ firewall acts as an intermediary, breaking the direct connection between client and server. **proxy/application**
22. A WAF (Web Application Firewall) operates at OSI Layer **7 or application**
23. A **DMZ** is a semi-trusted network zone between the internal network and the internet. **Demilitarized Zone**
24. A VPN creates a **virtual** private network over a public infrastructure.

Question Bank- Part 2

25. A Site-to-Site VPN that connects offices of the same company is called an _____-based VPN. **internal**
26. IPSec can operate in two modes: Transport mode and _____ mode. **tunneling**
27. WAP was designed to deal with the constraints of mobile devices, such as less bandwidth and **less** power.
28. The security protocol in the WAP stack is called **wtls**.
29. PGP provides confidentiality and _____ for email. **authentication**
30. S/MIME is a standard for secure _____. **e-mail**
31. The standard protocol for sending email is **smtp**.
32. The protocol for retrieving email while leaving copies on the server is **imap**.
33. In a monoalphabetic cipher, if 'A' maps to 'D', it will **always** map to 'D' throughout the text.
34. The set of all possible keys for a cipher is called the **key space**
35. A _____ attack tries every possible key to break a cipher. **brute-force**
36. A firewall packet can be Accepted, Denied, or **rejected**
37. UTM stands for Unified **threat** Management.
38. The WAP **gateway** translates between WAP protocols and Internet protocols.
39. WTLS is based on the Internet standard **tls**.
40. POP3, by default, **delete** email messages from the server after downloading.
-

Detailed Questions

1. What are the two primary goals of cryptography?
2. Explain the fundamental difference between symmetric and asymmetric encryption.
3. Describe how the Caesar Cipher works and its major weakness.
4. What is the difference between a block cipher and a stream cipher? Give an example of each.
5. Why is key length so important in symmetric cryptography?
6. What are the main advantages and disadvantages of symmetric-key cryptography?
7. What are the main advantages and disadvantages of asymmetric-key cryptography?
8. Explain the purpose of the Diffie-Hellman key exchange protocol.

Question Bank- Part 2

9. How does a hash function differ from encryption?
10. What is the purpose of an HMAC?
11. List and briefly describe three types of cryptanalysis attacks.
12. What is the difference between a firewall "whitelist" and "blacklist" approach?
13. Describe the function of a packet-filtering firewall. What are its limitations?
14. How does a stateful inspection firewall improve upon a basic packet filter?
15. What is the primary role of a proxy (application) firewall?
16. What specific threats does a Web Application Firewall (WAF) aim to prevent?
17. What additional capabilities does a Next-Generation Firewall (NGFW) have compared to a traditional stateful firewall?
18. What is the purpose of a DMZ in network architecture?
19. Compare and contrast Remote Access VPNs and Site-to-Site VPNs.
20. What is the function of a tunneling protocol in a VPN?
21. What were the main constraints that led to the development of WAP?
22. What is the role of the WAP Gateway?
23. Describe the two main services provided by PGP.
24. What is the key difference between how POP3 and IMAP handle email on the server?
25. How does Cipher Block Chaining (CBC) mode add security to block cipher operation?
26. Explain the concept of a "man-in-the-middle" attack in the context of key exchange.
27. What is the relationship between WTLS in WAP and TLS on the Internet?
28. Why might an organization choose to use a UTM appliance?
29. What does "egress filtering" on a firewall help to prevent?
30. How does public-key cryptography enable both encryption and digital signatures?